



Universität Hamburg

DER FORSCHUNG | DER LEHRE | DER BILDUNG

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Antrag auf Gewährung einer Reisebeihilfe vom 18.7.2024

Sehr geehrte Frau Dholakia,

ich freue mich, Ihnen mitteilen zu können, dass die Universität Ihnen für die Teilnahme an der Konferenz „IGU Konferenz“ vom 25. bis 30.8.2024 in Dublin eine Beihilfe bis zur Höhe von

EURO 600,-

aus Stiftungsmitteln gewährt.

Die bewilligten Gelder sollen dazu dienen, die in Ihrem Antrag aufgeführten Kosten für die Konferenzgebühr, die Flüge, die Transportkosten vor Ort und ausnahmsweise die Unterkunft zu finanzieren.

Bitte beachten Sie Folgendes: Die Bewilligung erfolgt unter der Voraussetzung, dass zum Zeitpunkt der Reise die aktuellen Reisebestimmungen der Universität Hamburg eine Reise an den Veranstaltungsort gestatten.

Für die Abrechnung der bewilligten Gelder nutzen Sie bitte den beiliegenden Abrechnungsbogen. Nach Beendigung der Maßnahme muss innerhalb von 3 Monaten eine Abrechnung mit Originalbelegen und einem Abschlussbericht bei mir erfolgen. Nach Ablauf der 3 monatigen Frist erlischt der Erstattungsanspruch. Für weitergehende Fragen stehe ich gern zur Verfügung.

Ich wünsche Ihnen für Ihr Vorhaben viel Erfolg und verbleibe

mit freundlichen Grüßen

gez.

Martina Jaeger



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FAKULTÄT
SCHOOL OF INTEGRATED CLIMATE AND
EARTH SYSTEM SCIENCES



Role of Livestock Enteric Methane in Climate Vulnerability in India (2003 – 2022) (Poster #3199)

Arnaz Dholakia¹ & Abhineety Goel²

Photo: pixabay/RonaldPlett

To examine the role and impact of cattle enteric methane on climate vulnerability.

Livestock contributes to 35–40% of methane emissions globally.

Agriculture is the predominant source of methane largely originating from livestock.

49.1% of enteric methane was contributed by cattle, 42.8% by buffaloes, 5.38% by goat and 2.59% by sheep[1]. The beef and dairy cattle industry is responsible for generating 14.5% of global greenhouse gas emissions[2]. During 1961–2010, the increase in CH₄ emissions (70.6%) from livestock population of India is much greater than the increase in CH₄ emissions from livestock population of world (54.3%)[1].

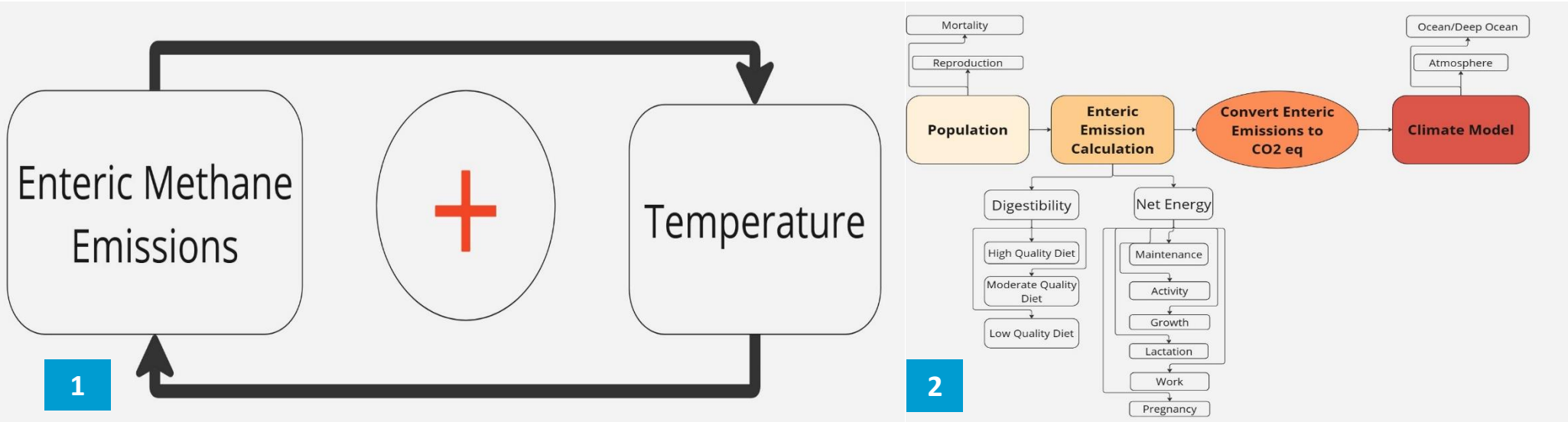
Cattle emit CH₄ through burping and flatulence[3]. Enteric fermentation occurs in the cow's rumen (first chamber of the stomach), where microbes break down feed, producing volatile fatty acids (VFAs) and gases like methane. VFAs provide energy for growth and milk production, while CH₄ is released through belching. Diet, feeding patterns, and physiological state influence VFA production and methane emissions.[4]

Different cattle breeds emit different quantities of CH₄ in different physiological stages. During 2014 – 15 and 2021 – 22, the CH₄ emissions from Indigenous Cattle grew by 14.25% and from Crossbred Cattle by 59.94%.

Climate change will affect livestock production significantly through their effects on feed production feed quality and composition, pest and disease dynamics, and animal health and production. Extreme events like droughts (Drought of 1987) affected 168 million cattle due to feed and fodder availability and water scarcity[1].

Methane is a potent GHG with 23 times the warming potential of Carbon dioxide.

CH₄ emissions and temperature creates a bidirectional feedback loop; a self-reinforcing cycle in which increased temperatures stimulate CH₄ and vice-versa.



- 1) Bidirectional feedback loop representation.
- 2) Model representation.

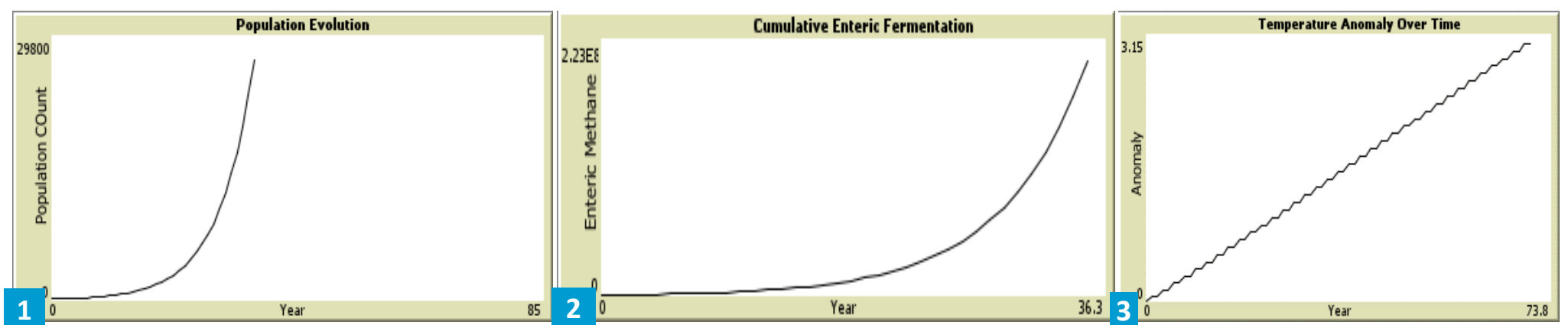
Diet Quality	SSP 1 – 1.9	SSP 1 – 2.6	SSP 2 – 4.5	SSP 3 – 7.0	SSP 5 – 8.5
High Quality	0.075 ± 0.0007	0.14 ± 0.0013	0.197 ± 0.0018	0.28 ± 0.0028	0.356 ± 0.004
Moderate Quality	0.081 ± 0.0007	0.152 ± 0.0013	0.212 ± 0.002	0.31 ± 0.003	0.383 ± 0.004
Low Quality	0.083 ± 0.0007	0.154 ± 0.0013	0.216 ± 0.002	0.315 ± 0.003	0.389 ± 0.004

- 1) Temperature anomaly (in C) under different SSP scenarios and qualities of diet generated using NetLogo 6.4.0 software upon running 1240 simulations.

Modelling Temperature Anomaly due to Enteric Methane Emissions.

The study first models the population, followed by modelling enteric emissions emitted under parameters – quality of diet, number of hours worked and net energy taken up by each cattle to perform different activities, and uses these emissions as CO₂ eq to estimate temperature anomaly under SSP scenarios. This approach establishes the first link between CH₄ emissions influencing the climate.

- The population is modelled at 60% reproductive rate and 15% mortality rate with the lifespan of 16 years and the purpose of dairy.
- This model projects the vulnerability caused due to cattle on the climate via temperature anomaly over 35 years.
- The bifurcation while modelling for enteric emission enables us to understand the net energy taken up by cattle and design policy prescriptions at a microscopic level.
- The feeding style – pasture, grazing and stall – have also been taken into consideration. Cattle that were fed via pastures emitted less methane compared to stall and grazing.
- The correlation between temperature anomaly under SSP scenarios and number of hours worked is 0.04. While the correlation is moderately strong, it indicates that over working the cattle increase its net energy for work and activity; therefore, increases the amount of CH₄ it emits.
- **Limitations:** The model assumes uniform diet, working hours, and general net energy across the entire population, limiting accuracy. No externalities are accounted for in the model.



- 1) Population model – first component of model.
- 2) Cumulative enteric methane emissions – second component of model.
- 3) Temperature anomaly due CO₂ eq of CH₄ emissions under SSP 5-8.5 – third component of model.

Recommendations.

Feeding Strategies:

- Provision of high quality diet with increased concentrates (concentrates >90).
- Provision of pasture feeds and rotational grazing.
- Increasing protein supplement in the diet.
- Using the technique of precision feeding in order to administer proportionate diets.
- Improve forage quality. High sugar ryegrass leads to a 7.5–21.0% increase in milk yield and a 7.1–25.7% decrease in excrement nitrogen. Re-seeding native grass species with those with higher productivity or C allocation to deeper roots, or introducing legumes into grazing lands, can all promote soil C in rangeland soils and reduce N emissions[1].
- Dietary inclusion of free oils or oil-rich feeds (e.g. oilseeds, distillers' grains, and micro-algae), biologically active plant compounds (e.g. condensed tannins, saponins, and essential oils), rumen fermentation modifiers (e.g. yeast and bacterial direct-fed microbials)[5].

Rumen Manipulation Strategies:

- Use of bacteriocins, ionophores, fats/oils, prebiotics and probiotics, halogenated methane analogues[1].

Population Modelling Strategies:

- Optimal livestock rearing.
- Improving breed management.
- Improving manure management.
- Administering methanogens vaccines to optimise VFAs.
- Regular health checkups.

General Recommendations :

- Creation of more intergovernmental policies and programs for creating mitigation strategies.

- [1] Thamil Vanan, T., Divya Lakshmi, D. (2020). Sustainable Livestock Management Systems for Indian Rural Livelihood: Mitigation of Climate Change. In: Venkatramanan, V., Shah, S., Prasad, R. (eds) Global Climate Change: Resilient and Smart Agriculture. Springer, Singapore.
- [2] Quinton, A. (2019, June 27). *Cows and climate change*. UC Davis.
- [3] Thakuri, S., Baskota, P., Khatri, S. B., Dhakal, A., Chaudhary, P., Rijal, K., & Byanju, R. M. (2020). Methane emission factors and carbon fluxes from enteric fermentation in cattle of Nepal Himalaya. *The Science of the Total Environment*.
- [4] Münger, A., & Kreuzer, M. (2006). Methane emission as determined in contrasting dairy cattle breeds over the reproduction cycle. *International Congress Series*, 1293, 119–122.
- [4] Gonzalez-Recio, O., Martínez-Álvarez, M., Tiezzi, F., Saborio-Montero, A., Maltecca, C., & Roehe, R. (2023). Invited Review: Novel methods and perspectives for modulating the rumen microbiome through selective breeding as a means to improve complex traits: implications for methane emissions in cattle. *Livestock Science*, 105171.
- [4] Meo-Filho, P., Hood, J., Lee, M., Fleming, H., Meethal, M., & Misselbrook, T. (2023). Performance and enteric methane emissions from housed beef cattle fed silage produced on pastures with different forage profiles. *Animal*, 100726.
- [4] Nylly, E., Prestløkken, E., Eknæs, M., Eikanger, K. S., Heldal Hagen, L., & Kidane, A. (2023). Inclusion of Red Macroalgae (*Asparagopsis taxiformis*) in Dairy Cow Diets Modulates Feed Intake, Chewing Activity and Estimated Saliva Secretion. *Animals*, 13(3), 489.
- [5] Beauchemin, K. A., McAllister, T. A., & McGinn, S. M. (2009). Dietary mitigation of enteric methane from cattle. *CABI Reviews*, 1–18.

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- 2) Assistant Professor at FLAME University, India.
- 3) Credits to Dr. Miguel Rodriguez at Universität Hamburg.





35TH INTERNATIONAL GEOGRAPHICAL CONGRESS 2024

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This is to certify that

Ms. Arnaz Nizar Dholakia

attended the 35th International Geographical Congress 2024

Dublin City University, Ireland

24 - 29 August 2024

Jackie McGann

IGC 2024 Project Manager

Role of Livestock Enteric Methane in Climate Vulnerability in India (2003 – 2022)

Abstract: Livestock such as cattle play a crucial role in global agriculture, particularly through the release of methane (CH₄) during enteric fermentation. Methane is a potent greenhouse gas with twenty-three times the warming potential of carbon dioxide (CO₂) (Dopelt, Radon and Davidovitch, 2019). This phenomenon can create a bidirectional feedback loop; a self-reinforcing cycle in which increased temperatures stimulate CH₄ production in livestock. Addressing this loop is critical to mitigating the impact of livestock emissions on climate change. This paper addresses the urgent need to assess the current and projected climate risks associated with enteric methane emissions from livestock such as cattle. Focusing on India between 2003 and 2022, the study uses arithmetic modelling to quantify these emissions at different physiological stages of cattle, using IPCC Tier 1 and 2 emission factors for spatial and temporal analysis. In particular, it incorporates Indian livestock census data, highlighting the significant CH₄ growth in the indigenous and crossbred dairy cattle population estimated to be 14.25% and 59.94~60% from 2014-15 to 2021-22. Globally, COP28 discussed methane and projected a 13% increment of CH₄ between 2023 and 2030 (United Nations Environmental Program, 2023). Furthermore, they urged the need for more data collection and public accessibility, thus planning for mitigation strategies. Thus, the study calls for creation of more intergovernmental policies and programs to mitigate increasing methane emissions

Introduction

Indian agriculture is a prominent occupation as it contributed 18.3% to the total economy in the financial year 2022-23 (Contribution of Agricultural Sector in GDP, 2023). India with its largest population (1.44 billion as in 2024), has the fifth global economy yet the poverty level rate in India is around 4.5-5%, most rural communities still depend on livestock. Dairy is the single largest agricultural commodity contributing 5 % of the Indian economy and employing more than 80 million farmers directly. The country ranks first in milk production contributing 23% of global milk production. Cattle, especially those used for beef and dairy production, are a significant contributor to global methane emissions, generating an estimated 4,623 million metric tons of carbon dioxide (CO₂) equivalent emissions annually (IPCC, 2018). Specifically, beef cattle are responsible for 2,495 million metric tons and dairy cattle for 2,128 million metric tons of CO₂ equivalent emissions per year (Garnett et al., 2014). The beef and dairy cattle industry is responsible for generating 14.5% of global greenhouse gas (GHG) emissions (The Global Methane Emission from Cows Issue, 2022, Cows and Climate Change, 2023). More than half of this gas is methane (CH₄) which traps heat in the atmosphere twenty-five times more effectively than carbon dioxide (The Global Methane Emission from Cows Issue, 2022, Cows and Climate Change, 2023). It was estimated that global GHG emissions was 49.4 billion tonnes of CO₂ equivalent of which CH₄ has 23-fold greater potential for global warming than CO₂ (Dopelt et al., 2019). One of the key characteristics of CH₄ is that it has the potential to recycle itself and enter the atmosphere as CO₂. Therefore, the gas that lived once in the atmosphere as CH₄, now has a higher probability to return to the atmosphere as CO₂. Thus, adding to the already existing CO₂ bank. It is rhetoric that there are various greenhouse gasses apart from methane. However, the factor that marks the difference between each of these gasses lies in their radiative factor unit.

Several variables impact methane emissions from cattle, including feed intake, carbohydrate type in the diet, feed processing, addition of lipids or ionophores to the diet, and ruminal

microbiota changes. Alteration of these elements has the potential to minimize methane emissions from cattle (Johnson & Johnson, 1995). Enteric methane emitted in kilogram (kg) per year per cattle head is calculated based on the estimates retrieved from IPCC Tier 1 and Tier 2 estimates. Tier 1 estimates entail utilizing the default enteric CH₄ emission factors, which offer set values for each species of animal in various parts of the world, regardless of changes in the physiological condition or output level of the animal. Alternatively, Tier 2 approaches more precisely represent country-specific data on livestock numbers, feed and feed consumption characteristics, country-specific emission factors are substituted for default emission factors. (IPCC, 2006, Azizi et al., 2017).

The Food Climate Research Network published a report in 2014 wherein they examined cattle, their grazing systems and how their activities contribute to the greenhouse gas emissions; and ultimately, to climate change. It is worth noting that the supply chain generated 7.1 Gt CO₂ equivalent of emissions, globally. The cattle dominate with 65% of the total emissions. Recently, scholars worldwide have acknowledged increasing detrimental potent effects of methane emissions largely emitted from cow belching. Thus, cows have become one of the major sources in the livestock stock industry that already contributes to 17% of the GHG global emissions.

In the undertaken study, the author proposes coupled model called Cattle Emission-Climate Interaction (CECI) model. The purpose of this study and the model is the estimate the changes in global temperature due to enteric methane emissions based on a modelled cattle population. This study acts as a prototype to further our understanding on the magnitude of cattle as a part of livestock management. While the case study is that of India, the model broadly represents cattle and enteric methane emission dynamics. The coefficients used in this study is tailored towards Indian cattle and therefore, the magnitude of cattle rearing in India versus rest of the world could be different. However, the study speculates this impact in a controlled manner and Indian cattle could be considered as a representative of cattle management in the Global South.

Literature Review

Enteric fermentation (also known as enteric methane emissions) occurs in the rumen - the first chamber of the cow's stomach, and involves the breakdown of ingested feed by microbes by producing Volatile Fatty Acids (VFA) and gasses such as methane and carbon dioxide. VFAs are absorbed by the walls of the rumen and transported to the liver when the animal utilizes it for energy. Bodily activities such as maintenance, growth and milk production are completed with the help of these VFAs. Factors such as the diet of the animal, feeding patterns and physiological state play an important role in the regulation of the VFA (Münger & Kreuzer, 2006, Beauchemin et al., 2009, Cows, Methane, and Climate Change, 2020, Gonzalez-Recio et al., 2023, Meo-Filho et al., 2023, Nyløyet al., 2023, Martínez-Marín et al., 2023). During enteric

fermentation, microbes in the rumen break down carbohydrates and fibre in the cow's diet, producing a mixture of gasses as a byproduct, including methane. The methane is then released from the cow's body through belching (Münger & Kreuzer, 2006, Beauchemin et al., 2009, Cows, Methane, and Climate Change, 2020, Gonzalez-Recio et al., 2023, Meo-Filho et al., 2023, Nyløyet al., 2023, Martínez Marín et al., 2023).

Fraser et al. (2014) first quantified methane emissions from free ranging beef cattle pastures on common grassland types in the lowlands and the uplands. They concluded that the effects of breed type were relatively small compared to combined influence of pasture type and location (Fraser et al., 2014). However, the study is based out of the UK and there are higher chances of geographical differences, grazing patterns, breed that define indigenous and exotic could be different. In Fraser et al's study (2014), it was claimed that the amount of methane produced per day was lower for cattle in the uplands as opposed to those who grazed in the lowlands. However, Thakuri et al study (2020) claimed that the estimated enteric emissions in the hilly region are not statistically significant to the emissions in the plain region; however significant differences exist between breeds and at different life stages.

Cattle emit methane through burping and flatulence; the quantity emitted by each cattle is (a) different from the other (b) differs based on their individual physiological life stage (c) differs based on the geographical location (Thakuri et al., 2020). Albeit, the authors claim that enteric methane of cattle in the plain region is approximately same to those cattle in hilly regions; i.e. there is not much of a statistical difference.

The physical processing of forages such as chopping, grinding, and steam treatment improves the forage digestibility and mitigates enteric methane production in ruminants (Grossi et al., 2018). The most crucial factor that affects the performance of dairy cows is their Dry Matter Intake (DMI). Dry matter is the amount of feed consumed (kg) after it has been corrected for the water content in the complete diet. For example, consumption of 10 kg of a diet that contains 70 dry matters would result in a dry matter intake (DMI) of 7 kg. Feed Intake is positively related to animal size, growth rate, milk production and pregnancy.

Dall-Orsoletta et al (2019) estimated the emissions through a decade of base data under the criteria of breed, feeding strategy, first calving age. The results generated from their study showed a decrease in enteric methane emissions as it did not factor population changes through different physiological stages. To understand the factors affecting production, such as the DMI, intake pattern, and feed efficiency, behavioral assessments like eating and rumination activities are commonly used. The behavior of animals varies in response to alterations in their diet. Feed digestibility (DE%) is the portion of gross energy (GE) in the feed that is not excreted in the feces. For ruminants, common ranges for feed digestibility are 45 - 55% for crop by-product and range lands; 55 - 75% for good pastures, good preserved forages and grain supplemented forage-based diets and 75 - 85% for grain-based diets fed in feedlots. The digestibility of forages decreases with increasing maturity and is typically lowest during the dry season. Digestibility, intake and growth are co-dependent phenomenon. Feeds with high DE% will often result in a higher feed intake and increased growth. The use of unconventional feeding ingredients might result in changes in behavior that can impact an animal's intake, health, and overall performance (Nyløyet al., 2023). Similarly, Johnson and Johnson (1995) and Garnett, et al., (2014) also emphasize on the role of diet for enteric methane emissions.

The cattle require net energies for different physiological and physical activities that it carries out in its daily life. Some of these activities include maintenance, activity, growth, lactation, work and pregnancy. Net energy for maintenance is defined as the amount of energy needed to keep the animal in equilibrium where body energy is neither gained nor lost. The coefficients used for calculating the net energy for maintenance is 0.386 KJ/day kg for lactating and 0.322 MJ/day kg for non-lactating cattle and increasing the feed use for maintenance is associated

with greater methane emissions. This energy is obtained from the quality of food that the cattle consume. Rhetorically, higher the diet quality, lower the enteric methane emissions. Therefore, high quality of diet, is a unit of MJ per kilograms of Dry Matter (MJ/kg DM), which includes vegetative legumes and grasses, is defined as a range between 7.5 – 8.5 (MJ/kg DM), moderate quality of diet includes mild seasonal legumes and grasses and is defined as a range between 5.5 – 6.5 (MJ/kg DM) and low-quality diet includes straws and matured grasses and is defined as a range between 3.5 – 5.5 (MJ/kg DM).

The net energy for activity is the product of the net energy taken up for maintenance and feeding style. The cattle are fed in three distinct styles – stall¹, grazing² and pasture³ for which the coefficients are 0, 0.36 and 0.17 respectively. The feeding styles are a dimensionless variable.

The net energy for growth is defined of Mega Joules of energy per day and is converted from calories. This energy takes into account the average live body weight, coefficient based on the animal, mature live body weight and average daily weight gain in the unit of kg per day.

The net energy for lactation also has a unit of Mega Joules per day and is expressed as a function of the amount of milk that is produced by the population and its fat content. Normally, the fat content of milk produced by cattle is 4%

The cattle take up net energy to work. This can be condensed to the notion of the level of strenuousness of the work performed by the animal that has an influence on the energy it consumes. The undertaken study follows the guidelines stated by Bamulian and Kartiarso of accounting for 10% of the days net energy for maintenance is required per hour for typical work for cattle. Furthermore, the gestation period, therefore net energy for pregnancy, for cattle is 281 days and accounts for 10% of net energy for maintenance.

Upon accounting for the aforementioned diversification of net energy taken up by an individual animal, the ratio of net energy in diet for maintenance to digestible energy consumed and ratio of net energy available for growth to digestible energy consumed is calculated so as to calculate the gross energy taken up by cattle each day in Mega Joules. This gross energy is further used to calculate emission factor based on different permutations of feed quality and feeding styles (also known as feed lots).

Model Methodology

The undertaken study generated a coupled model called Cattle Emission-Climate Interaction (CECI) model. CECI assumes that all cattle are under fed under domestic management for dairy purposes with a life span of 16 years. The physiological aspects (reproductivity) is akin to a cow irrespective of its breed; therefore, the model assumes that all cattle are of one breed and are not subjected to differentiation of species.

CECI is based on a (hypothetical) population model which is then used as a source of input for the enteric emission model. It calculates the emissions from the population model and plugs the CO₂ equivalent into a climate model with two feedback parameters (ocean/deep ocean and

¹ Animals are confined to a small area (eg: tethered, barn, pen) with the result that they expend very little or no energy to acquire food.

² Animals graze in open range land or hilly terrain and expend significant energy to acquire food.

³ Animals are confined in areas with sufficient forage requiring modest energy expense to acquire food.

atmosphere). The primary agent in the model is therefore, cow. The cow has five functions – to reproduce, to sustain, to expend energy by undertaking various activities, to produce milk and mortality. The model assumes to have a male counterpart; however, the enteric model does not account for its (male) population evolution and subsequent enteric emissions.

The population model has some percentage for reproductivity, mortality and consumption of food. The enteric model accounts for net energy for maintenance, activity, growth, lactation, work, and pregnancy. This model then calculates the feed digestibility based on the feeding style (pasture, stall or grazing) and quality of diet (high, moderate, low). It then calculates gross energy intake based on the following formula

$$GE = \frac{\left\{ \left[\frac{NE_m + NE_a + NE_L + NE_{work} + NE_P}{REM} \right] + \left[\frac{NE_G + NE_{milk}}{REG} \right] \right\}}{\frac{DE\%}{100}}$$

Upon calculating the GE, the model calculates a methane emission factor (EF) based on feedlots and energy content of methane. The EF, which is calculated as GgCH₄ per year (Gigagram of methane per year) based on the population, is then calculated as CO₂ equivalent to be plugged into the climate model.

The climate model is a simple dynamic representation of how emissions influence atmospheric carbon concentration and temperature, while accounting for both effects and feedback mechanisms.

$$F^\circ = E \quad \text{eq 1}$$

Eq 1 represents direct relationship between Emissions (E) and Radiative Forcing (F). Radiative forcing is a measure of the energy imbalance in the Earth's atmosphere due to greenhouse gas emissions. In this model, the radiative forcing is assumed to be directly proportional to emissions. This simplification helps in focusing on the primary drivers of climate change—greenhouse gases like CO₂—without delving into the complexities of different types of forcing (e.g., aerosols, land use changes).

$$\dot{C} = \beta E + BF - \sigma C \quad \text{eq 2}$$

Eq 2 describes the dynamics of atmospheric carbon concentration (Cdot). The term βE captures how emissions contribute to the concentration of carbon in the atmosphere. The coefficient β represents the efficiency with which emissions translate into increased carbon concentration. For example, a higher β would imply that a larger proportion of emissions remain in the atmosphere, possibly due to lower absorption by oceans or terrestrial sinks. The term BF represents feedback mechanisms related to forcing. Positive feedbacks (e.g., the release of methane from cattle) can amplify the effects of emissions on carbon concentration. These feedbacks are essential in climate modelling because they can significantly alter the trajectory of atmospheric carbon levels over time.

The term $-\sigma C$ accounts for the natural decay of carbon from the atmosphere, which can occur through processes like absorption by oceans and vegetation. The coefficient σ represents the rate at which carbon is removed from the atmosphere. A higher σ would imply a faster removal rate, which would mitigate the rise in carbon concentration.

$$\dot{T} = \mu \ln(c) - \alpha T \quad \text{eq 3}$$

The third equation governs the temperature change $\dot{T}$ as a function of the logarithm of the relative carbon concentration ccc , defined as the ratio of the current atmospheric carbon concentration C to the pre-industrial level $C_{\text{pre-industrial}}$. This logarithmic relationship reflects the well-established principle that the radiative forcing effect of CO_2 increases logarithmically with its concentration. The coefficient μ determines the sensitivity of temperature to changes in carbon concentration. This is akin to the concept of climate sensitivity, which is a measure of how much global temperatures are expected to rise in response to increased greenhouse gas concentration. The term $-\alpha T$ represents the decay or reduction in temperature change over time. This could be due to various processes such as radiative cooling, heat uptake by oceans, or other climate system responses that dampen temperature increases. The coefficient α controls the strength of this damping effect, with higher values indicating a more rapid return to equilibrium temperatures after an initial disturbance.

CECI produces temperature anomaly from the emissions from cattle for 35 years. The temperature anomaly is calculated using two ways – (1) using Shared Socioeconomic Pathways (SSP) Scenarios (2) adjusting different feedback parameters so as to regulate and investigate the implications of controlled heating and cooling effects in the climate system. Climate system is not dynamic and SSP scenarios in CECI run under specific values for feedback parameter. Therefore, approach (2) provides leeway for the user to adjust and manipulate and perhaps explore perturbations in the climate system itself.

Two of the most important aspects of the enteric modelling lies in the dietary choices provided to the bovine. Therefore, incorporating the feeding styles and diet qualities provides information for specific policy prescription.

All models are wrong but some are useful. It must be acknowledged that the CECI has limitations as it does not involve any spatiality and the preference of feeding and dietary quality is uniform and user dependent. Furthermore, the model does not involve any environmental, social or economic externalities that could affect the decisions of the provider (user).

Results

The aim of the undertaken study was to assess the temperature evolution from an initial temperature of 15 degree Celsius due to enteric CH_4 emissions of cattle. The population is modelled at 60% reproductive rate and 15% mortality rate with the lifespan of 16 years and the animal is deployed for dairy purposes. This model projects the vulnerability caused due to cattle on the climate system via temperature anomaly over 35 years. The model has run for a total of 1240 simulations and the results have been generated as observed in in Table 1.

Diet Quality	SSP 1 – 1.9	SSP 1 – 2.6	SSP 2 – 4.5	SSP 3 – 7.0	SSP 5 – 8.5
High Quality	0.075 0.0007	$\pm 0.14 \pm$ 0.0013	$0.197 \pm$ 0.0018	$0.28 \pm$ 0.0028	$0.356 \pm$ 0.004
Moderate Quality	0.081 0.0007	$\pm 0.152 \pm$ 0.0013	$0.212 \pm$ 0.002	$0.31 \pm$ 0.003	$0.383 \pm$ 0.004
Low Quality	0.083 0.0007	$\pm 0.154 \pm$ 0.0013	$0.216 \pm$ 0.002	$0.315 \pm$ 0.003	$0.389 \pm$ 0.004

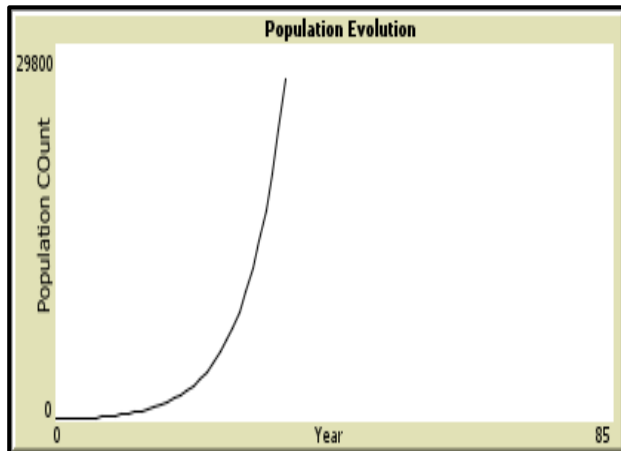
Table 1: Temperature anomalies generated based on the quality of diet across different SSP scenarios at 95% confidence interval.

Climate change will affect livestock production significantly through their effects on feed production feed quality and composition, pest and disease dynamics, and animal health and production.

IPCC AR 6 highlights the impact of extreme events on food availability and its implications on food prices, globally. In 1987, extreme event of drought affected 168 million cattle in India due to feed and fodder availability and water scarcity. Furthermore, drought induces temperature increments which furthers our hypothesis of positive bidirectional feedback loop created between CH₄ and temperature. Morignat et al (2015) studied the relationship between risk of death and temperature in cattle in France and concluded that the risk of death increases below a cold threshold (4.4% - 5% decrease below 1 degree C Cold Threshold) and above a hot temperature threshold (4.6% - 5.6% above 1 degree C Hot Threshold). This result is marginally indistinguishable between beef and dairy cattle. Similarly, results generated from CECI, while does not include extreme events, however, predicts that the temperature threshold devoid of extreme events under all SSP scenarios is at 0.1 degree Celsius. One must be cognizant that the climate system in this model is simple and does not involve various feedback parameters. Nevertheless, this estimate can provide preliminary direction of the temperature anomaly. Image 3 provides with this understanding. The temperature curve is incremental step wise. This highlights the notion that the climate model stabilizes during its run. The magnitude of the temperature anomaly is different under different SSP scenarios; however, the direction of the curve is same overall. Image 3 is from SSP 5 – 8.5 which at the end of 73 years suggests that the temperature anomaly would be close to 3.15 degree celsius.

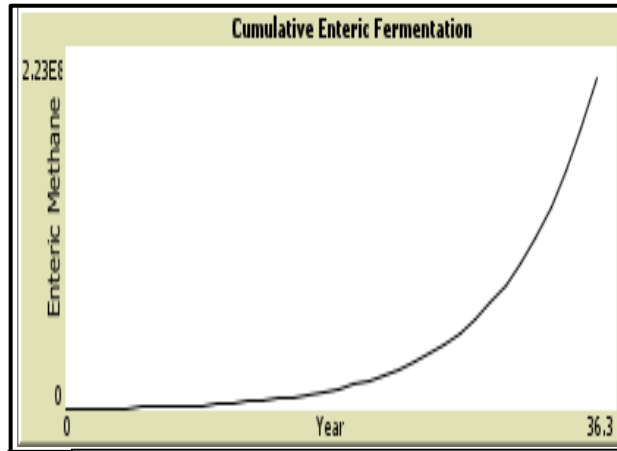
Quality of the diet fed to the cattle is an important function of the temperature anomaly caused due to cattle CH₄ emission. Under each SSP scenario, the model experiments with different feed qualities and feeding styles. This experimentation gives better understanding when comparing low quality diet with high quality diet. Among the three feeding styles, it was observed that cattle that were fed via pastures emitted less methane compared to stall and

grazing -fed cattle. Furthermore, the number of hours that cattle worked played an important role in the emission bank. The correlation between number of hours worked and the emissions is estimated to be 0.4. The correlation is moderately strong and indicates that exhausting the cattle requires higher net energy intake and thus increases the enteric emission of the cattle.



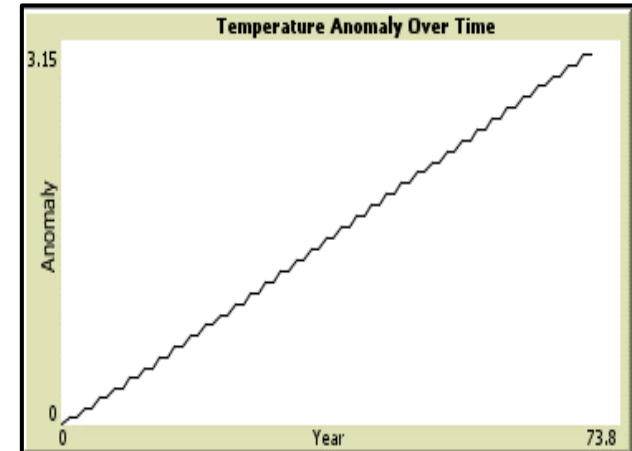
1

Image 1: Population Evolution in 85 years



2

Image 2: Cumulative Enteric Emission of CH₄



3

Image 3: Temperature Anomaly due to Enteric CH₄ Emissions

Recommendations

The undertaken study suggests recommendations on the bases of feeding, rumen manipulation, and population modelling. It is empirical that further studies need to be conducted to gain further insights in CH₄ emissions from cattle. Moving further on from academic and scientific research, it is considerable that we encourage the creation of more intergovernmental policies and programs for creating mitigation strategies.

The notable study by Beauchemin et al., (2009) suggests further improvement in the cattle diet. This can be achieved through a variety of approaches, such as improving the quality and digestibility of feed, reducing the amount of feed wasted, and implementing management practices that improve animal health and productivity. One of the mitigation strategies taken up by the Global North is providing seaweed to the cattle. The anthropogenic benefit of seaweed consumption is the reduced methane emissions. Irish and UK seaweeds have proven that they are not only capable of reducing production but also providing immunity.

Dietary inclusion of free oils or oil-rich feeds (e.g. oilseeds, distillers' grains, and micro-algae), biologically active plant compounds (e.g. condensed tannins, saponins, and essential oils), rumen fermentation modifiers (e.g. yeast and bacterial direct-fed microbials), and forage quality improvements may allow for further reductions in cattle methane emissions. Improving higher forage quality would be highly influential. High sugar ryegrass leads to a 7.5–21.0% increase in milk yield and a 7.1–25.7% decrease in excrement nitrogen. Re-seeding native grass species with those with higher productivity or Carbon allocation to deeper roots, or introducing legumes into grazing lands, can all promote soil Carbon in rangeland soils and reduce Nitrogen emissions (Eggleston et al., 2006)

The best nutritional plan will be determined by the farmer itself along with provision of pastoral or rotational grazing feeding styles. The role of the farmer is to undertake precision feeding in order to administer proportionate diets. On the other hand, incorporating protein supplements and provision of high-quality diet with increased concentrates (concentrates >90) would be helpful in controlling enteric emissions.

Rumen manipulation techniques would primarily include the use of bacteriocins, ionophores, fats/oils, prebiotics and probiotics, halogenated methane analogues. It must be noted that the undertaken study does not account for the administration of medications or supplements.

Furthermore, prospective mitigation methods must be evaluated from the standpoint of the whole life cycle, since a decrease in GHG emissions in one sector of the production cycle can frequently lead to changes in GHG emissions in another (Beauchemin et al., 2009).

The Global North has internalized, emphasized and prioritized health care provision for cattle, Global South is yet to come to power with the former in its management techniques for optimal livestock rearing, breed management, manure management and regular health checkup. From a global perspective, a serious consideration to administering methanogens vaccines to optimise VFAs should be taken.

Lastly, the nexus of cows (or cattle), climate change, soil, and atmosphere are empirical as land management practices have a higher probability of exchanging carbon and nitrogen between soil and atmosphere (Rivera, Chara, Barhona, 2018). Thus, identification of the sources and sinks of these greenhouse gasses opens doors to research in the domain of terrestrial

ecosystems. Humidity, temperature, aeration, excreta composition and fertilizer that are applied to soil have a decisive influence over soil greenhouse gas emissions.

References

- Azizi, A. A., Mardhati, M., Mohd Fairuz, M., Shadatul Azdawiyah, A. T., Nurul Ain, A. N., Fauzi, J., Mohamad Hariz, A. R., & Resigner, A. (2017). Local emission factors estimates for methane emission from cattle enteric fermentation using IPCC tier-2 methodology. *Malaysian Society of Animal Production*, 20(2). https://www.researchgate.net/publication/326109676_Local_emission_factors_estimates_for_methane_emission_from_cattle_enteric_fermentation_using_IPCC_tier_2_methodology
- Bamualim, A., & Kartiarso. (1985). Nutrition of draught animals with special reference to Indonesia. In J. W. Copland (Ed.), *Draught animal power for production* (ACIAR Proceedings Series No. 10). Australian Centre for International Agricultural Research.
- Beauchemin, K. A., McAllister, T. A., & McGinn, S. M. (2009). Dietary mitigation of enteric methane from cattle. *CABI Reviews*, 1–18.
- Garnett, T., Godde, C., Roeoes, E., Smith, P., Boer, I. J. M., Ermgassen, E., Herrero, M., Middelhaar, C. E., Schader, C., & Zanten, H. H. E. (2014). Grazed and confused? Ruminating on cattle, grazing systems, methane, nitrous oxide, the soil carbon sequestration question – and what it all means for greenhouse gas emissions. *Wageningen University & Research*. <https://library.wur.nl/WebQuery/wurpubs/fulltext/427016>
- Gonzalez-Recio, O., Martínez-Álvaro, M., Tiezzi, F., Saborío-Montero, A., Maltecca, C., & Roehe, R. (2023). Invited review: Novel methods and perspectives for modulating the rumen microbiome through selective breeding as a means to improve complex traits: Implications for methane emissions in cattle. *Livestock Science*, 105171. <https://doi.org/10.1016/j.livsci.2023.105171>
- Intergovernmental Panel on Climate Change (IPCC). (2006). *Guidelines for national greenhouse gas inventories* (H. S. Eggleston, L. Buendia, K. Miwa, T. Ngara, & K. Tanabe, Eds.). <https://www.osti.gov/etdeweb/biblio/20880391>
- Intergovernmental Panel on Climate Change (IPCC). (2018). Summary for policymakers. In V. Masson-Delmotte, P. Zhai, H.-O. Pörtner, D. Roberts, J. Skea, P. R. Shukla, A. Pirani, W. Moufouma-Okia, C. Péan, R. Pidcock, S. Connors, J. B. R. Matthews, Y. Chen, X. Zhou, M. I. Gomis, E. Lonnoy, T. Maycock, M. Tignor, & T. Waterfield (Eds.), *Global warming of 1.5°C. An IPCC special report on the impacts of global warming of 1.5°C above pre-industrial levels and related global greenhouse gas emission pathways, in the context of strengthening the global response to the threat of climate change, sustainable development, and efforts to eradicate poverty* (pp. 3-24). Cambridge University Press. <https://doi.org/10.1017/9781009157940.001>
- Johnson, K. A., & Johnson, D. E. (1995). Methane emissions from cattle. *Journal of Animal Science*, 73(8), 2483–2492. <https://doi.org/10.2527/1995.7382483x>
- Martínez-Marín, G., Schiavon, S., Tagliapietra, F., Cecchinato, A., Toledo-Alvarado, H., & Bittante, G. (2023). Interactions among breed, farm intensiveness and cow productivity on

predicted enteric methane emissions at the population level. *Italian Journal of Animal Science*, 22(1), 59–75. <https://doi.org/10.1080/1828051x.2022.2158953>

Meo-Filho, P., Hood, J., Lee, M., Fleming, H., Meethal, M., & Misselbrook, T. (2023). Performance and enteric methane emissions from housed beef cattle fed silage produced on pastures with different forage profiles. *Animal*, 100726. <https://doi.org/10.1016/j.animal.2023.100726>

Morignat, E., Gay, E., Vinard, J., Calavas, D., & Hénau, V. (2015). Quantifying the influence of ambient temperature on dairy and beef cattle mortality in France from a time-series analysis. *Environmental Research*, 140, 524–534. <https://doi.org/10.1016/j.envres.2015.05.001>

Münger, A., & Kreuzer, M. (2006). Methane emission as determined in contrasting dairy cattle breeds over the reproduction cycle. *International Congress Series*, 1293, 119–122.

Nylø, E., Prestløkken, E., Eknæs, M., Eikanger, K. S., Heldal Hagen, L., & Kidane, A. (2023). Inclusion of red macroalgae (*Asparagopsis taxiformis*) in dairy cow diets modulates feed intake, chewing activity and estimated saliva secretion. *Animals*, 13(3), 489.

Quinton, A. (2019, June 27). Cows and climate change. *UC Davis*.

Thakuri, S., Baskota, P., Khatri, S. B., Dhakal, A., Chaudhary, P., Rijal, K., & Byanju, R. M. (2020). Methane emission factors and carbon fluxes from enteric fermentation in cattle of Nepal Himalaya. *The Science of the Total Environment*. <https://doi.org/10.1016/j.scitotenv.2020.142569>

Thamil Vanan, T., & Divya Lakshmi, D. (2020). Sustainable livestock management systems for Indian rural livelihood: Mitigation of climate change. In V. Venkatramanan, S. Shah, & R. Prasad (Eds.), *Global climate change: Resilient and smart agriculture*. Springer.

United Nations Environmental Program (2023) *COP28: Methane pledge by the ‘giants behind the climate crisis’ falls short, says Guterres*. <https://news.un.org/en/story/2023/12/1144282> (Accessed: January 12, 2024).